

Question Number	Error	Line Number	Error Type	Error Correction
1	int choice	25	Syntax Error	Missing semicolon at the end of the declaration: int choice;
2	default	65	Syntax Error	A switch label needs a colon. Use: default:
3	if(totalBooks > 20)	90	Logical Error (Incorrect Bounds Check)	books[] has size 20 (valid indices 0-19), but the check allows totalBooks to reach 20 before flagging "Library Full," causing an out-of-bounds write. Check before writing and use: if(totalBooks >= 20) { printf("Library Full\n"); return; }
4	for(i=0;i<=totalBooks;i++)	100	Logical Error (Off-by-One)	Using <= accesses one element past the last valid book. Use: for(i=0;i<totalBooks;i++)
5	books[id].price = books[id].price - 10;	123	Runtime Error (Unvalidated Array Index)	id comes directly from user input and is never checked against valid bounds before use. Validate first: if(id >= 0 && id < totalBooks) books[id].price -= 10;
6	books[id].price = books[id].price + 10;	132	Runtime Error (Unvalidated Array Index)	id comes directly from user input and is never checked against valid bounds before use. Validate first: if(id >= 0 && id < totalBooks) books[id].price += 10;
7	totalBook	173	Semantic Error (Undeclared Identifier)	Typo: the actual variable is "totalBooks". Correct it to: totalBooks
8	calculateFine();	177	Semantic Error (Undeclared/Undefined Function)	calculateFine() is never declared or defined anywhere in the file. Add a prototype and a matching function definition, or remove the call.
9	b.bookId = "101";	184	Type Mismatch Error	bookId is an int; cannot assign a string literal. Use: b.bookId = 101;
10	b.price = "500";	190	Type Mismatch Error	price is a float; cannot assign a string literal. Use: b.price = 500;
11	if(title == books[0].title)	201	Logical Error	This compares array addresses, not the string contents. Use: if(strcmp(title, books[0].title) == 0)
12	books[id] = NULL;	213	Compile Error (Type Mismatch)	books[id] is a struct, not a pointer; NULL cannot be assigned to it. Also id is never bounds-checked. Reset the struct instead, e.g. memset(&books[id], 0, sizeof(struct Book)); after validating id.
13	books[id].price = books[id].price + 100;	222	Runtime Error (Unvalidated Array Index)	id comes directly from user input and is never checked against valid bounds before use. Validate first: if(id >= 0 && id < totalBooks) books[id].price += 100;
14	avg = calculateAveragePrice();	230	Semantic Error (Undeclared/Undefined Function)	calculateAveragePrice() is never declared or defined anywhere in the file. Define it, or remove/replace the call.
15	fp = fopen("library.txt", "r");	239	Logical Error (Wrong File Mode)	The file is opened in read-only mode ("r") but then written to with fprintf. Open it for writing instead: fp = fopen("library.txt", "w"); and check fp != NULL before using it.
16	fp = fopen("missing.txt", "r");	250	Runtime Error (Missing NULL Check)	If the file does not exist, fopen returns NULL, and the following fscanf(fp,...) call will crash. Add a check: if(fp == NULL) { printf("File not found\n"); return; }
17	*ptr = 100;	280	Runtime Error (Null Pointer Dereference)	ptr is NULL; dereferencing it crashes the program. Allocate memory first, e.g. ptr = malloc(sizeof(int));
18	arr[10] = 200;	284	Runtime Error (Array Index Out of Bounds)	arr has size 5 (valid indices 0-4). Use a valid index, e.g. arr[4] = 200;
19	printf("%d\n", a/b);	289	Runtime Error (Division by Zero)	b is 0, causing undefined behavior/crash. Check the divisor before dividing: if(b != 0) printf("%d\n", a/b);
20	strcpy(name, "Programming");	296	Runtime Error (Buffer Overflow)	name is only 5 bytes, but "Programming" needs 12 bytes (11 chars + null terminator). Enlarge the buffer: char name[20]; or use strncpy with a size limit.
21	ptr = malloc(5);	306	Runtime Error (Insufficient Memory Allocation)	Only 5 bytes are allocated, but the code writes 20 ints (80 bytes) into ptr, causing a heap buffer overflow. Allocate enough space: ptr = malloc(20 * sizeof(int));
22	printf("%d\n", ptr[0]);	315	Runtime Error (Use After Free)	ptr was already freed on the line above; accessing it afterward is undefined behavior. Read the value before calling free(ptr), or remove this access.
23	free(ptr);	317	Runtime Error (Double Free)	ptr was already freed earlier in this function; freeing it again is undefined behavior and can corrupt the heap. Remove this duplicate free(ptr) call.